9.2 Air Microbiology

Earth's environment is endowed with atmosphere, hydrosphere, lithosphere and biosphere. In the 1930s, **F.C. Meier** coined the term "Aerobiology". Air is a natural resource and is fundamental to human life as it makes breathing possible. Its universal presence and requirement for the survival of human beings make air an important environmental factor. Air contains significant number of microorganisms. It also acts as a medium for the transmission of microorganisms including bacteria, viruses, fungi, yeast and protozoa. Air borne transmission is one of the important modes for the transmission of infectious diseases. Aeromicrobiology is the study of air borne microorganisms.

Infobits

Miquel (1883) and Carnally & colleagues (1887) carried out the most systematic studies in airborne microorganisms.

9.2.1 Layers of Atmosphere

Earth's atmosphere is divided into five major layers (Figure 9.1) they are:

- Exosphere : 700 to 10,000 km
- Thermosphere : 80 to 700 km
- Mesosphere : 50 to 80 km
- Stratosphere : 12 to 50 km
- Troposphere : 0 to 12 km

The air in the troposphere, the layer close to the earth is constantly in motion and the temperature decreases with increasing altitude, reaching a low of -57°C at the apex of this region. Microorganisms are frequently found in the lower portion of the troposphere, where they are dispersed by the air currents. Most of the microbes present in the troposphere are either spore formers or microbes that are easily dispersed in air. The stratosphere has a temperature range of -80°C to -10°C. The temperature in the stratosphere can reach a maximum of several thousand degrees. Microorganisms are not found in the upper regions of the atmosphere because of the temperature extremes, scarcity of available oxygen, absence of nutrients and moisture, and low atmospheric pressure. The relatively low humidity in the atmosphere (especially during daylight hours) and UV rays from the sun, limit the types and number of microorganisms that are able to survive in this part of the biosphere.

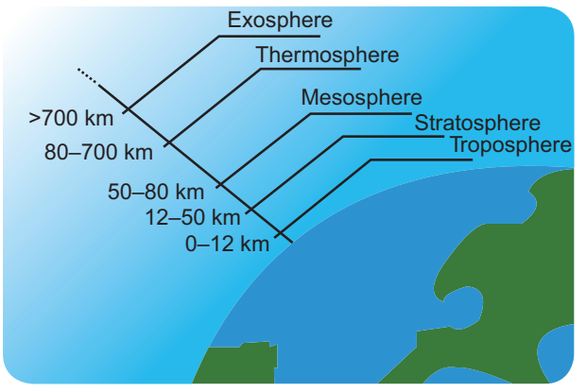

Figure 9.1: Diagram showing layers of atmosphere

9.2.2 Composition of Air

The air in our atmosphere is composed of different gaseous molecules. The most common gases are nitrogen (78%), CO_2 (0.034%) oxygen (21%) argon (1%) and other molecules in trace level are present in the atmosphere.

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